Math 54 — Week 3

Linear independence; linear transformations; the matrix of a linear transformation (Lay 1.7–1.9)

§1 Facts used from Weeks 1–2

(W1) A consistent system has exactly one solution if it has no free variables, infinitely many otherwise. Free variables correspond to non-pivot columns.

(W2) $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$. $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b}$ is a combination of the columns.

(W3) The columns of an $m \times n$ matrix A span $\mathbb{R}^m \iff A$ has a pivot in every row.

(W4) $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ and $A(c\mathbf{u}) = cA\mathbf{u}$.

(W5) $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\iff$ it has a free variable.

§2 This week

§2.1 Linear independence (Lay 1.7)

The question. In Week 2, $\mathbf{a}_3 = 2\mathbf{a}_1 - \mathbf{a}_2$, so $\mathbf{a}_3$ added nothing to the span of $\mathbf{a}_1, \mathbf{a}_2$. How do we detect in general that some vectors in a list are redundant?

Definition. $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation $x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$ has only the trivial solution. It is **linearly dependent** if there are weights $c_1, \ldots, c_p$, *not all zero*, with $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$; such an equation is a **linear dependence relation**.

- **Columns of a matrix.** The equation $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$ is $A\mathbf{x} = \mathbf{0}$. So the columns of A are linearly independent $\iff$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff$ no free variables $\iff$ **a pivot in every column** (W5).

Worked example. $\mathbf{v}_1 = (1, 2, 3)$, $\mathbf{v}_2 = (4, 5, 6)$, $\mathbf{v}_3 = (2, 1, 0)$.

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x_3 is free, so the set is dependent. Taking $x_3 = 1$: $x_1 = 2$, $x_2 = -1$. Dependence relation: $2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$. Check: $(2, 4, 6) - (4, 5, 6) + (2, 1, 0) = (0, 0, 0)$.

Small sets.

- One vector: $\{\mathbf{v}\}$ is independent $\iff$ $\mathbf{v} \neq \mathbf{0}$ (since $x\mathbf{v} = \mathbf{0}$ with $x \neq 0$ forces $\mathbf{v} = \mathbf{0}$).
- Two vectors: $\{\mathbf{v}_1, \mathbf{v}_2\}$ is dependent $\iff$ one is a multiple of the other. (If $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$ with $c_1 \neq 0$, then $\mathbf{v}_1 = -\frac{c_2}{c_1}\mathbf{v}_2$; and conversely.) Geometrically: dependent exactly when they lie on one line through $\mathbf{0}$.

Theorem (Dependence means one vector is a combination of the others)

$S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ with $p \geq 2$ is linearly dependent $\iff$ at least one vector in S is a linear combination of the others. In fact, if S is dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ with $j > 1$ is a linear combination of the *preceding* vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

The idea

A dependence relation is an equation in which some vector has a nonzero weight, and any such vector can be solved for. To get the stronger statement, choose the *last* vector with a nonzero weight: then everything else in the relation comes before it.

Proof. 1. If $\mathbf{v}_j = c_1\mathbf{v}_1 + \ldots$ (the others), move $\mathbf{v}_j$ to the other side: a relation with weight $-1 \neq 0$ on $\mathbf{v}_j$. So S is dependent.

2. Conversely, let $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ with not all $c_i = 0$, and let j be the largest index with $c_j \neq 0$. If $j = 1$, then $c_1\mathbf{v}_1 = \mathbf{0}$, so $\mathbf{v}_1 = \mathbf{0}$, excluded. So $j > 1$, and $\mathbf{v}_j = -\frac{1}{c_j}(c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1})$.

□

The theorem does *not* say every vector in a dependent set is a combination of the others: in $\{(1,0),(2,0),(0,1)\}$, $(0,1)$ is not.

Theorem (Too many vectors are dependent)

If $p > n$, any set of p vectors in $\mathbb{R}^n$ is linearly dependent.

Proof. The matrix $A = [\,\mathbf{v}_1 \cdots \mathbf{v}_p\,]$ is $n \times p$. It has at most n pivots (one per row), and $p > n$ columns, so some column has no pivot: a free variable in $A\mathbf{x} = \mathbf{0}$. □

Theorem (A set containing $\mathbf{0}$ is dependent)

If $\mathbf{v}_1 = \mathbf{0}$, then $1 \cdot \mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_p = \mathbf{0}$ is a dependence relation.

Geometry in $\mathbb{R}^3$. Three vectors are dependent exactly when they lie in a common plane through $\mathbf{0}$ (or on a line): by the theorem above, one of them lies in the span of the others.

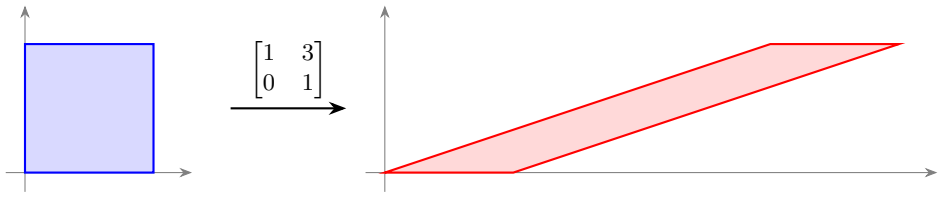
§2.2 Linear transformations (Lay 1.8)

The question. Instead of solving $A\mathbf{x} = \mathbf{b}$ for one $\mathbf{b}$, think of $\mathbf{x} \mapsto A\mathbf{x}$ as a function that moves every vector of $\mathbb{R}^n$. What kinds of functions arise, and which properties do they share?

Definition. A **transformation** $T\colon \mathbb{R}^n \to \mathbb{R}^m$ assigns to each $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$, the **image** of $\mathbf{x}$. $\mathbb{R}^n$ is the **domain**, $\mathbb{R}^m$ the **codomain**, and the set of all images is the **range**. A **matrix transformation** is $T(\mathbf{x}) = A\mathbf{x}$ for an $m \times n$ matrix A; its range is the span of the columns of A.

Definition. T is **linear** if $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u}, \mathbf{v}$ in the domain and all scalars c.

- Every matrix transformation is linear (W4).
- **Consequences.** $T(\mathbf{0}) = \mathbf{0}$ (take $c = 0$). $T(c\mathbf{u}+d\mathbf{v}) = cT(\mathbf{u})+dT(\mathbf{v})$, and more generally $T(c_1\mathbf{v}_1+\cdots+c_p\mathbf{v}_p) = c_1T(\mathbf{v}_1)+\cdots+c_pT(\mathbf{v}_p)$ (superposition).
- **Quick test for non-linearity**: if $T(\mathbf{0}) \neq \mathbf{0}$, T is not linear. The translation $T(\mathbf{x}) = \mathbf{x} + \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is not linear.
- Examples on $\mathbb{R}^2$: $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ is a *shear* (slides points horizontally by $3\times$ their height); $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ *projects* onto the x_1-axis; rI is a *dilation* ($r > 1$) or *contraction* ($0 \leq r < 1$).



unit square (scaled) image: a parallelogram

A shear. Each point (x_1, x_2) moves to $(x_1 + 3x_2,\ x_2)$: the bottom edge stays put and the top edge slides right. Lines go to lines and the origin stays fixed, as for every linear transformation.

§2.3 The matrix of a linear transformation (Lay 1.9)

The question. Is every linear transformation $\mathbb{R}^n \to \mathbb{R}^m$ a matrix transformation? If so, how do we find the matrix from a description such as "rotate by 30°"?

Theorem (The standard matrix)

Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be linear. Then there is exactly one matrix A with $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$, namely

$$A = [\,T(\mathbf{e}_1)\ \ T(\mathbf{e}_2)\ \ \cdots\ \ T(\mathbf{e}_n)\,],$$

where $\mathbf{e}_j$ is the j-th column of I_n. A is the **standard matrix** of T.

The idea

Every $\mathbf{x}$ is a combination of the standard vectors: $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$. A linear T respects combinations, so $T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n)$. The right side is exactly the matrix with columns $T(\mathbf{e}_j)$ times $\mathbf{x}$. So knowing where T sends the n standard vectors determines T everywhere.

Proof. 1. $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$, so by linearity $T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n) = A\mathbf{x}$ with A as stated.
2. Uniqueness: if $T(\mathbf{x}) = B\mathbf{x}$ for all $\mathbf{x}$, then column j of B is $B\mathbf{e}_j = T(\mathbf{e}_j)$, the same as column j of A.

□

Rotation. T rotates $\mathbb{R}^2$ counterclockwise by φ. Then $\mathbf{e}_1 = (1, 0)$ goes to $(\cos\varphi, \sin\varphi)$ and $\mathbf{e}_2 = (0, 1)$ goes to $(-\sin\varphi, \cos\varphi)$:

$$A = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}.$$

Other standard matrices on $\mathbb{R}^2$, found the same way (where do $\mathbf{e}_1$ and $\mathbf{e}_2$ go?):

move	$T(\mathbf{e}_1),\ T(\mathbf{e}_2)$	matrix
reflection through the x_1-axis	$(1,0),\ (0,-1)$	$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$
reflection through the x_2-axis	$(-1,0),\ (0,1)$	$\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$
reflection through the line $x_2 = x_1$	$(0,1),\ (1,0)$	$\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$
reflection through the origin	$(-1,0),\ (0,-1)$	$\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$
horizontal shear by k	$(1,0),\ (k,1)$	$\begin{bmatrix}1 & k\\ 0 & 1\end{bmatrix}$
projection onto the x_2-axis	$(0,0),\ (0,1)$	$\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$

There is no need to memorise the table: each row takes a few seconds to re-derive.

Existence and uniqueness questions for T.

Definition. $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$. It is **one-to-one** if each $\mathbf{b}$ is the image of *at most one* $\mathbf{x}$.

Theorem (One-to-one means only $\mathbf{0}$ goes to $\mathbf{0}$)
A linear T is one-to-one $\iff$ $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Proof. 1. If T is one-to-one: $T(\mathbf{0}) = \mathbf{0}$, so $\mathbf{0}$ is the only vector sent to $\mathbf{0}$.
2. If only $\mathbf{0}$ goes to $\mathbf{0}$, and $T(\mathbf{u}) = T(\mathbf{v})$, then $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$, so $\mathbf{u} - \mathbf{v} = \mathbf{0}$.

□

Theorem (Onto and one-to-one from the standard matrix)
Let A be the standard matrix of T. (a) T is onto $\iff$ the columns of A span $\mathbb{R}^m$ $\iff$ A has a pivot in every row. (b) T is one-to-one $\iff$ the columns of A are linearly independent $\iff$ A has a pivot in every column.

Proof. (a) Onto says $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$; use (W3). (b) By the previous theorem, one-to-one says $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, which is independence of the columns, which is a pivot in every column. □

- If $n > m$ (more columns than rows), T is never one-to-one. If $n < m$, T is never onto.

Shortcuts

- **Independence by inspection**: more vectors than entries ⇒ dependent; contains $\mathbf{0}$ ⇒ dependent; two vectors ⇒ check for a multiple. Otherwise row reduce $[\,\mathbf{v}_1 \cdots \mathbf{v}_p\,]$ and look for a pivot in every column.
- **Finding a dependence relation**: solve $A\mathbf{x} = \mathbf{0}$, set one free variable to 1 and the others to 0.
- **Standard matrix**: column j is $T(\mathbf{e}_j)$. For compositions of geometric moves, just track $\mathbf{e}_1$ and $\mathbf{e}_2$ through the moves.
- **Is T linear?** First check $T(\mathbf{0}) = \mathbf{0}$. Formulas with constants added, products of variables, $|x|$, or squares are not linear.
- **Pivot in every row** = onto = columns span; **pivot in every column** = one-to-one = columns independent. One row reduction of A answers both.
- **Given $T(\mathbf{u})$ and $T(\mathbf{v})$**, the image of $c\mathbf{u} + d\mathbf{v}$ is $cT(\mathbf{u}) + dT(\mathbf{v})$; no matrix needed.

Traps

- Independence is about the homogeneous equation; whether $A\mathbf{x} = \mathbf{b}$ is solvable for some $\mathbf{b}$ is a different question.
- In a dependent set, not every vector is a combination of the others.
- Three vectors in $\mathbb{R}^3$, no two of which are multiples, can still be dependent (all in one plane).
- "Onto" and "one-to-one" use different pivot conditions (rows versus columns); do not mix them.
- The range of T is a subset of the codomain; they are equal only when T is onto.
- Rotation matrices: the minus sign is in the top right for counterclockwise rotation.

§3 Homework this week

HW2, Section 1.7: 1, 6, 10, 13, 21, 24, 26, 33, 36, 38. Section 1.8: 2, 5, 7, 11, 17, 24, 26, 29, 31. Section 1.9: 2, 6, 13, 15, 19, 21, 23, 25, 32, 33. HW3, Section 1.8: 41, 44.

§4 Problem set

P1. Are the columns of $\begin{bmatrix} 1 & -3 & 2 \\ 3 & -8 & 5 \\ -2 & 7 & -3 \end{bmatrix}$ linearly independent?

P2. For which h are $(1, -1, 4)$, $(3, -5, 7)$, $(-1, 5, h)$ linearly dependent? For that h, give a dependence relation.

P3. Decide by inspection, with a reason: (a) $(1, 2), (3, 1), (0, 5)$; (b) $(2, -4, 6), (-1, 2, -3)$; (c) $(1, 0, 2), (0, 0, 0), (5, 1, 1)$; (d) $(1, 2, 0), (0, 1, 3)$.

P4. $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ first reflects through the x_1-axis and then rotates counterclockwise by $90°$. Find its standard matrix, and describe T as a single geometric move.

P5. Let $T(\mathbf{x}) = A\mathbf{x}$ with $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 2 & -3 & 5 \end{bmatrix}$. Find all $\mathbf{x}$ with $T(\mathbf{x}) = (-1, 2, 0)$. Is T one-to-one? Onto?

P6. Is $T(x_1, x_2) = (x_1 + x_2,\ 2x_1 - 1)$ linear? Is $S(x_1, x_2) = (x_2,\ 3x_1 - x_2,\ 0)$?

P7. $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ has standard matrix $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 5 & 1 & 7 \\ 1 & 3 & 2 & 5 \end{bmatrix}$. Is T onto? One-to-one?

P8. $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ is linear with $T(\mathbf{e}_1) = (2, -1, 3)$ and $T(\mathbf{e}_2) = (0, 4, 1)$. Find $T(3, -2)$ and the standard matrix of T.